COMPUTER NETWORKS- LAB ASSIGNMENT 9

NAME: KESHAV. G

REG. NO: 24BCE1003

SLOT: L57+L58

A small organization has three departments located in different buildings. Each department has its own Local Area Network (LAN). The networks are connected through three routers.

Your task is to design and configure the network so that all PCs in different departments can communicate with each other.

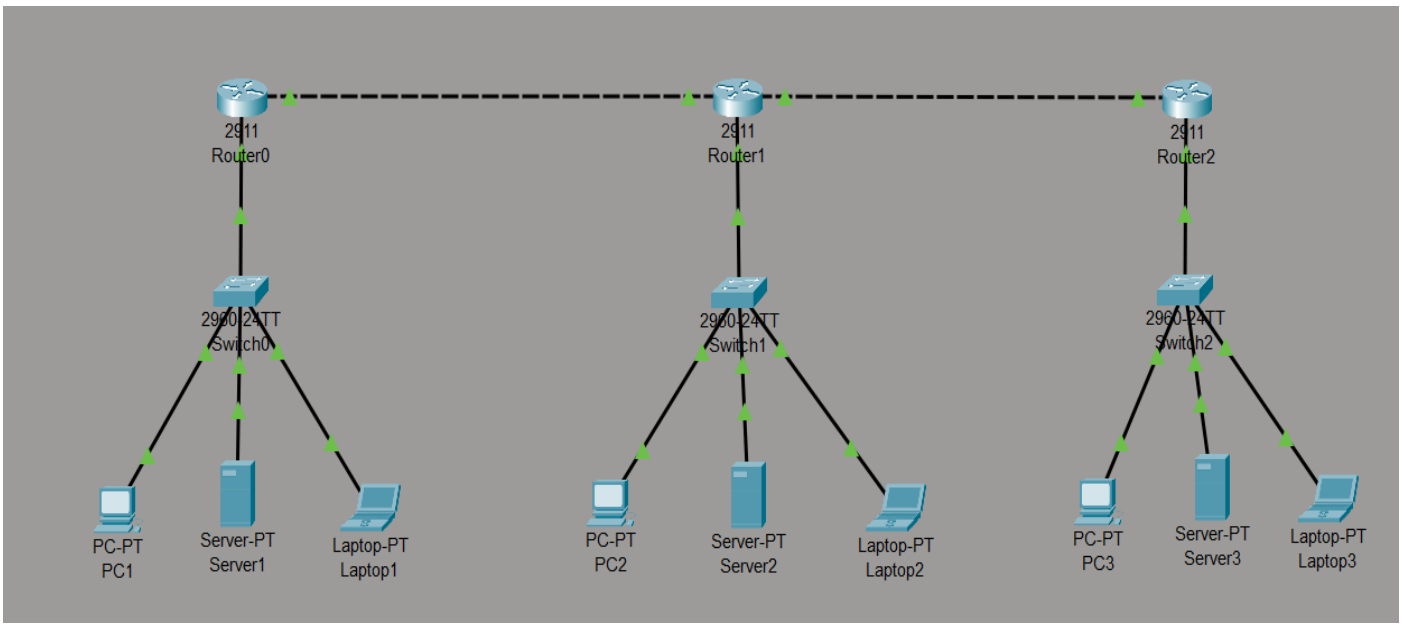
Devices Required:

- 3 Routers (Router0, Router1, Router2)
- 3 Switches
- One PC, one server, one laptop for each LAN (Naming convention can be PC1, Laptop1, server1 for LAN1; PC2, Laptop2, server2 for LAN2;...)

i) Once the configuration is done, verify the packet transmission is failure or success.
ii) Implement access control mechanisms for the following cases:

- PC in LAN 1 cannot transmit packets to server in LAN 3
- Laptop in LAN2 cannot transmit packets to PC in LAN1
- Server in LAN 1 cannot transmit packets to Laptop in LAN 3

Topology:



Routers used : 2911 (2910 Routers have only 2 GigabitEthernet ports, but this topology requires 3 ports for the router, hence 2911 Router was chosen)

Switches used: 2960

End devices used: PC, Server, Laptop

Routing tables:

Switch0:

DEVICE	IP ADDRESS	GATEWAY
PC1	**192.168.1.2**	**192.168.1.1**
Server1	**192.168.1.3**	**192.168.1.1**
Laptop1	**192.168.1.4**	**192.168.1.1**
Router0 GigabitEthernet 0/0	**192.168.1.1**	**-----**

Switch1:

DEVICE	IP ADDRESS	GATEWAY
PC2	**192.168.2.2**	**192.168.2.1**
Server2	**192.168.2.3**	**192.168.2.1**
Laptop2	**192.168.2.4**	**192.168.2.1**
Router1 GigabitEthernet 0/1	**192.168.2.1**	**-----**

Switch2:

DEVICE	IP ADDRESS	GATEWAY
PC3	**192.168.3.2**	**192.168.3.1**
Server3	**192.168.3.3**	**192.168.3.1**
Laptop3	**192.168.3.4**	**192.168.3.1**
Router2 GigabitEthernet 0/1	**192.168.3.1**	**-----**

WAN Connection (Router0->Router1)

DEVICE	IP ADDRESS
Router0 GigabitEthernet 0/1	**10.0.0.1/30**
Router1 GigabitEthernet 0/0	**10.0.0.2/30**

Router1->Router2

DEVICE	IP ADDRESS
Router1 GigabitEthernet 0/2	**10.0.0.5/30**
Router2 GigabitEthernet 0/0	**10.0.0.6/30**

Packet transmission verification:

i) PC1->PC2

Simulation panel:

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC1	ICMP
	0.001	PC1	Switch0	ICMP
	0.002	Switch0	Router0	ICMP
	0.003	Router0	Router1	ICMP
	0.004	Router1	Switch1	ICMP
	0.005	Switch1	PC2	ICMP
	0.006	PC2	Switch1	ICMP
	0.007	Switch1	Router1	ICMP
	0.008	Router1	Router0	ICMP
	0.009	Router0	Switch0	ICMP
	0.010	Switch0	PC1	ICMP

Status:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC2	ICMP		0.000	N	0	(e...	(del...

ii) PC2->PC3:

Simulation panel:

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC2	ICMP
	0.001	PC2	Switch1	ICMP
	0.002	Switch1	Router1	ICMP
	0.003	Router1	Router2	ICMP
	0.004	Router2	Switch2	ICMP
	0.005	Switch2	PC3	ICMP
	0.006	PC3	Switch2	ICMP
	0.007	Switch2	Router2	ICMP
	0.008	Router2	Router1	ICMP
	0.009	Router1	Switch1	ICMP
	0.010	Switch1	PC2	ICMP

Status:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC2	PC3	ICMP		0.000	N	0	(e...	(del...

iii) PC3->PC1:

Simulation panel:

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC3	ICMP
	0.001	PC3	Switch2	ICMP
	0.002	Switch2	Router2	ICMP
	0.003	Router2	Router1	ICMP
	0.004	Router1	Router0	ICMP
	0.005	Router0	Switch0	ICMP
	0.006	Switch0	PC1	ICMP
	0.007	PC1	Switch0	ICMP
	0.008	Switch0	Router0	ICMP
	0.009	Router0	Router1	ICMP
	0.010	Router1	Router2	ICMP
	0.011	Router2	Switch2	ICMP

Status:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC3	PC1	ICMP		0.000	N	0	(e...	(del...

Therefore, the connections have been established successfully between all 3 LANs, now we move on to implementing ACLs

ii) Access Control Lists:

- **On Router0:** Block PC1->Server3 and Server1->Laptop3

CLI Commands:

en

conf t

access-list 100 deny ip host 192.168.1.10 host 192.168.3.12

access-list 100 deny ip host 192.168.1.12 host 192.168.3.11

access-list 100 permit ip any any

interface g0/0

ip access-group 100 in

ctrl+c

exit

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 100 deny ip host 192.168.1.2 host 192.168.3.3
Router(config)#access-list 100 deny ip host 192.168.1.3 host 192.168.3.4
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/0
Router(config-if)#ip access-group 100 in
Router(config-if)#
Router#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

en

show access-lists

```
Router>en
Router#show access-lists
Extended IP access list 100
    10 deny ip host 192.168.1.2 host 192.168.3.3
    20 deny ip host 192.168.1.3 host 192.168.3.4
    30 permit ip any any
```

- **On Router1:** Block Laptop2->PC1

CLI commands:

en

conf t

access-list 101 deny ip host 192.168.2.11 host 192.168.1.10

access-list 101 permit ip any any

interface g0/1

ip access-group 101 in

exit

exit

ctrl+c

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 101 deny ip host 192.168.2.4 host 192.168.1.2
Router(config)#access-list 101 permit ip any any
Router(config)#interface g0/1
Router(config-if)#ip access-group 101 in
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

show access-lists

```
Router#show access-lists
Extended IP access list 101
    10 deny ip host 192.168.2.4 host 192.168.1.2
    20 permit ip any any
```

- **Router2:** Need not be configured as we have already added the necessary ACL blocking IP addresses in both Router0 and Router1 already

Demonstration:

Simulation Panel:

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC1	ICMP
	0.000	--	Server1	ICMP
	0.000	--	Server1	ARP
	0.000	--	Laptop2	ICMP
	0.000	--	Laptop2	ARP
	0.001	PC1	Switch0	ICMP
	0.001	Server1	Switch0	ARP
	0.001	Laptop2	Switch1	ARP
	0.002	Switch0	Router0	ICMP
	0.002	--	Switch0	ARP
	0.002	Switch0	PC1	ARP
	0.002	Switch0	Laptop1	ARP
	0.002	Switch1	PC2	ARP
	0.002	Switch1	Server2	ARP
	0.002	Switch1	Router1	ARP
	0.002	--	Router0	ICMP
	0.003	Switch0	Router0	ARP
	0.003	Router0	Switch0	ICMP
	0.003	Router1	Switch1	ARP
	0.004	Router0	Switch0	ARP
	0.004	Switch0	PC1	ICMP
	0.004	Switch1	Laptop2	ARP
	0.004	--	Laptop2	ICMP
	0.005	Switch0	Server1	ARP
	0.005	Laptop2	Switch1	ICMP
	0.005	--	Server1	ICMP
	0.006	Server1	Switch0	ICMP
	0.006	Switch1	Router1	ICMP
	0.006	--	Router1	ICMP
	0.007	Switch0	Router0	ICMP
	0.007	Router1	Switch1	ICMP
	0.007	--	Router0	ICMP
	0.008	Router0	Switch0	ICMP
	0.008	Switch1	Laptop2	ICMP

0.007	Router1	Switch1	ICMP
0.007	--	Router0	ICMP
0.008	Router0	Switch0	ICMP
0.008	Switch1	Laptop2	ICMP
0.009	Switch0	Server1	ICMP

Status:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC1	Server3	ICMP		0.000	N	0	(e...	(del...
	Failed	Server1	Laptop3	ICMP		0.000	N	1	(e...	(del...
	Failed	Lapto...	PC1	ICMP		0.000	N	2	(e...	(del...

Therefore, the required Access Control Lists have been implemented successfully to block specific packet transfers in the given network.